

Format of LSC Statement Packet

2017 - 11 - 27

Packet Size : 12 Byte ~ (12+[N]) Byte

Header 4 byte	Destination Address 2 byte	Source Address 2 byte	Statement 2 byte	Para Length 1 byte	Parameters Length [N] byte	Checksum 1 byte
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Header	Header1	0xFF
	Header2	0xFF
	Header3	0x4C
	Header4	0x53

Destination Address	Controller	Addr H	Each Addr 1~65535	BroadCast 0
		Addr L		
Source Address	Host	Addr H		
		Addr L		

Statement	Statement1	Type
	Statement2	Code

Length	Para Length	Length [N]	※ Depends on Statement
Para	Parameters	[N] byte	

Checksum	= SrcAddr H + SrcAddr L + Statement1 + Statement2 + Length (+ Para1 + Para... + ParaN)
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Type	DATA GET	0x10
	DATA SET	0x20
	EXECUTION	0x30
	Files Management	0x40

Response Data

DATA GET	Type	Code	Para Length	Para1	Para2	Para3	Para4	Para5	Para6	Para7	Para8	Para9	...	
Get LSC Version	0x10	1	0											
			3	HW Version	FW Major	FW Minor								
Get BAT Voltage		2	0											
			2	BAT VoltageH	BAT VoltageL									
Get Module ID (only 1EA Connected)		3	0											
			1	LSM ID										
Get Nbr of LSM(Connected)		4	1	Last ID										
			1	NumberOfLSM										
Get LSMs NowPosition		5	[N]LSMs	LSM ID_1st	...	LSM ID_[N-1]th	LSM ID_[N]th							
			[N]LSMs x 3	ID_1st	ID_1st PosH	ID_1st PosL	...	ID_[N]th	ID_[N]th PosH	ID_[N]th PosL				
Get LSMs NowVoltage		6	[N]LSMs	LSM ID_1st	...	LSM ID_[N-1]th	LSM ID_[N]th							
			[N]LSMs x 3	ID_1st	ID_1st VoltH	ID_1st VoltL	...	ID_[N]th	ID_[N]th VoltH	ID_[N]th VoltL				
Get LSMs NowCurrent		7	[N]LSMs	LSM ID_1st	...	LSM ID_[N-1]th	LSM ID_[N]th							
			[N]LSMs x 3	ID_1st	ID_1st NowCurrH	ID_1st NowCurrL	...	ID_[N]th	ID_[N]th NowCurrH	ID_[N]th NowCurrL				
Get LSMs LimitCurrent		8	[N]LSMs	LSM ID_1st	...	LSM ID_[N-1]th	LSM ID_[N]th							
			[N]LSMs x 3	ID_1st	ID_1st LmCurrH	ID_1st LmCurrL	...	ID_[N]th	ID_[N]th LmCurrH	ID_[N]th LmCurrL				
Get LSMs NowTemperature		9	[N]LSMs	LSM ID_1st	...	LSM ID_[N-1]th	LSM ID_[N]th							
			[N]LSMs x 2	ID_1st	ID_1st NowTemp	...	ID_[N]th	ID_[N]th NowTemp						
Get LSMs LimtTemperature		10	[N]LSMs	LSM ID_1st	...	LSM ID_[N-1]th	LSM ID_[N]th							
			[N]LSMs x 2	ID_1st	ID_1st LmtTemp	...	ID_[N]th	ID_[N]th LmtTemp						
Get LSMs GainP		11	[N]LSMs	LSM ID_1st	...	LSM ID_[N-1]th	LSM ID_[N]th							
			[N]LSMs x 2	ID_1st	ID_1st GainP	...	ID_[N]th	ID_[N]th GainP						
Get LSMs GainI		12	[N]LSMs	LSM ID_1st	...	LSM ID_[N-1]th	LSM ID_[N]th							
			[N]LSMs x 2	ID_1st	ID_1st GainI	...	ID_[N]th	ID_[N]th GainI						
Get LSMs GainD	13	[N]LSMs	LSM ID_1st	...	LSM ID_[N-1]th	LSM ID_[N]th								
		[N]LSMs x 2	ID_1st	ID_1st GainD	...	ID_[N]th	ID_[N]th GainD							
Get LSMs ControlMargin	14	[N]LSMs	LSM ID_1st	...	LSM ID_[N-1]th	LSM ID_[N]th								

Statement	Get LSMs ControlMargin		14	[N]LSMs x 3	ID_1st	ID_1st CM H	ID_1st CM L	...	ID_[N]th	ID_[N]th CM H	ID_[N]th CM L			
	Get LSMs FwdControl Value		15	[N]LSMs	LSM ID_1st	...	LSM ID_[N-1]th	LSM ID_[N]th						
				[N]LSMs x 3	ID_1st	ID_1st FCV H	ID_1st FCV L	...	ID_[N]th	ID_[N]th FCV H	ID_[N]th FCV L			
	Get LSMs BoundaryMax		16	[N]LSMs	LSM ID_1st	...	LSM ID_[N-1]th	LSM ID_[N]th						
				[N]LSMs x 3	ID_1st	ID_1st BNDMX H	ID_1st BNDMX L	...	ID_[N]th	ID_[N]th BNDMX H	ID_[N]th BNDMX L			
	Get LSMs BoundaryMin		17	[N]LSMs	LSM ID_1st	...	LSM ID_[N-1]th	LSM ID_[N]th						
				[N]LSMs x 3	ID_1st	ID_1st BNDMN H	ID_1st BNDMN L	...	ID_[N]th	ID_[N]th BNDMN H	ID_[N]th BNDMN L			
	Get LSMs LED Value		18	[N]LSMs	LSM ID_1st	...	LSM ID_[N-1]th	LSM ID_[N]th						
				[N]LSMs x 4	LSM ID_1st	ID_1st LED R	ID_1st LED G	ID_1st LED B	...	LSM ID_[N]th	ID_[N]th LED R	ID_[N]th LED G	ID_[N]th LED B	
	Get LSMs ModelVersion	19	[N]LSMs	LSM ID_1st	...	LSM ID_[N-1]th	LSM ID_[N]th							
			[N]LSMs x 3	ID_1st	ID_1st MV H	ID_1st MV L	...	ID_[N]th	ID_[N]th MV H	ID_[N]th MV L				
	Get LSMs FW Version	20	[N]LSMs	LSM ID_1st	...	LSM ID_[N-1]th	LSM ID_[N]th							
			[N]LSMs x 3	ID_1st	ID_1st FV H	ID_1st FV L	...	ID_[N]th	ID_[N]th FV H	ID_[N]th FV L				
	Get LSMs ZeroCompValue	21	[N]LSMs	LSM ID_1st	...	LSM ID_[N-1]th	LSM ID_[N]th							
		[N]LSMs x 3	ID_1st	ID_1st ZCPM H	ID_1st ZCMP L	...	ID_[N]th	ID_[N]th ZCMP H	ID_[N]th ZCMP L					
	Get Sync Info	22	1	넘어질때모션번호										
			2	모션번호 포즈번호										
< DATA GET > Process		Direction \ Packet		Header	Dest Addr	Src Addr	Statement	Length	Parameter N	CheckSum				
		HOST >> LSC (Command)		Header	LSC Addr	HOST Addr	Statement	ParaLength	Para	CheckSum				
		LSC >> HOST (Response)		Header	HOST Addr	LSC Addr	Statement	RespParaLength	RespPara	CheckSum				
DATA SET		Type	Code	Para Length	Para1	Para2	Para3	Para4	Para5	Para6	Para7	Para8	Para9	...
Set LSC Version		0x20	1	3	HW Version	FW Major	FW Minor							
Set LSC Address			2	2	Address H	Address L								
Set LSM Module ID			3	2	LSM ID	new LSM ID								
Set Sync Target Position			4	[N]LSMs x 4	LSM SyncID_1st	ID_1st SyncT/S	ID_1st SyncPosH	ID_1st SyncPosL	...	LSM SyncID_[N]th	ID_[N]th SyncT/S	ID_[N]th SyncPosH	ID_[N]th SyncPosL	
Set Sync LED Control Value			5	[N]LSMs x 4	LSM SyncID_1st	ID_1st LEDpreR	ID_1st LEDpreG	ID_1st LEDpreB	...	LSM SyncID_[N]th	ID_[N]th LEDpreR	ID_[N]th LEDpreG	ID_[N]th LEDpreB	
Set LSM Limit Current			6	[N]LSMs x 3	ID_1st	ID_1st LmCurrH	ID_1st LmCurrL	...	ID_[N]th	ID_[N]th LmCurrH	ID_[N]th LmCurrL			
Set LSM Limit Temperature			7	[N]LSMs x 2	ID_1st	ID_1st LmtTemp	...	ID_[N]th	ID_[N]th LmtTemp					
Set LSM GainP			8	[N]LSMs x 2	ID_1st	ID_1st GainP	...	ID_[N]th	ID_[N]th GainP					
Set LSM GainI			9	[N]LSMs x 2	ID_1st	ID_1st GainI	...	ID_[N]th	ID_[N]th GainI					
Set LSM GainD			10	[N]LSMs x 2	ID_1st	ID_1st GainD	...	ID_[N]th	ID_[N]th GainD					
Set LSM Control Margin			11	[N]LSMs x 3	ID_1st	ID_1st CM H	ID_1st CM L	...	ID_[N]th	ID_[N]th CM H	ID_[N]th CM L			
Set LSM ForwardControl Value			12	[N]LSMs x 3	ID_1st	ID_1st FCV H	ID_1st FCV L	...	ID_[N]th	ID_[N]th FCV H	ID_[N]th FCV L			
Set LSM Boundary Max			13	[N]LSMs x 3	ID_1st	ID_1st BNDMX H	ID_1st BNDMX L	...	ID_[N]th	ID_[N]th BNDMX H	ID_[N]th BNDMX L			
Set LSM Boundary Min			14	[N]LSMs x 3	ID_1st	ID_1st BNDMN H	ID_1st BNDMN L	...	ID_[N]th	ID_[N]th BNDMN H	ID_[N]th BNDMN L			
Set LSM Comm Baudrate			15	[N]LSMs x 4	ID_1st	ID_1st BPS H	ID_1st BPS M	ID_1st BPS L	...	ID_[N]th	ID_[N]th BPS H	ID_[N]th BPS M	ID_[N]th BPS L	
Set LSM ZeroComp Value			16	[N]LSMs x 3	ID_1st	ID_1st ZCPM H	ID_1st ZCMP L	...	ID_[N]th	ID_[N]th ZCMP H	ID_[N]th ZCMP L			
Set LSM Factory Reset			17	4	LSM ID	0x53	0x46	0x52						
Set LSS Module ID			18	3	Product ID	LSS ID	new LSS ID							
EXECUTION		Type	Code	Para Length	Para1	Para2	Para3	Para4	Para5	Para6	Para7	Para8	Para9	...
Exe LSM PWR ON/OFF			1	1	0:OFF / 1:ON									
Exe LSM Configure Save			2	[N]LSMs	LSM ID_1st	...	LSM ID_[N]th							
Exe LSM PosMove Torq			3	[N]LSMs x 4	LSM ID_1st	ID_1st Torque	ID_1st PosH	ID_1st PosL	...	LSM ID_[N]th	ID_[N]th Torque	ID_[N]th PosH	ID_[N]th PosL	
Exe LSM PosMove Speed			4	[N]LSMs x 4	LSM ID_1st	ID_1st Speed	ID_1st PosH	ID_1st PosL	...	LSM ID_[N]th	ID_[N]th Speed	ID_[N]th PosH	ID_[N]th PosL	
Exe LSM LED Control			5	[N]LSMs x 4	LSM ID_1st	ID_1st LED R	ID_1st LED G	ID_1st LED B	...	LSM ID_[N]th	ID_[N]th LED R	ID_[N]th LED G	ID_[N]th LED B	
Exe LSM Sync PosMove			6	[N]LSMs x 2	LSM ID_1st	ID_1st SyncMode	...	LSM ID_[N]th	ID_[N]th SyncMode					

	Exe LSM Sync LED Control	0x30	7	[N]LSMs	LSM ID_1st ... LSM ID_[N]th									
	Exe LSM Integration Control		8	[N]LSMs x 7	LSM ID_1st	ID_1st Torque	ID_1st PosH	ID_1st PosL	ID_1st LED R	ID_1st LED G	ID_1st LED B	...		
	Exe LSM Wheel Mode		9	[N]LSMs x 3	LSM ID_1st	ID_1st Dir(1: CW/2: CCW)	ID_1st WSpd(%)	...	LSM ID_[N]th	ID_[N]th Dir(1: CW/2: CCW)	ID_[N]th WSpd(%)			
	Exe LSM ActiveStop		10	[N]LSMs	LSM ID_1st	... LSM ID_[N]th								
	Exe LSM PassiveStop		11	[N]LSMs	LSM ID_1st	... LSM ID_[N]th								
	Exe Motion		12	3	MotionNumber	Start delay [ms]	Speed(-50~50)[%]							
	Exe Process		13	1	0:OFF / 1:ON									
	Exe Remocon		14	4	mode	0:OFF / 1:ON		L data	R data					
	Exe Motion info		15	2	MotionNumber	PosNumber								
	Files Management	Type	Code	Para Length	Para1	Para2	Para3	Para4	Para5	Para6	Para7	Para8	Para9	...
	Write PROCESS	0x40	1	6	PlatformH	PlatformL	FileSize 1	FileSize 2	FileSize3	FileSize4				

Examples

Exe Motion

0xFF	0xFF	0x4C	0x53	0x00	0x00	0x00	0x00	0x30	0x0C	0x03	MotionNumber	0x00	0x00	Checksum
								type	code	para length				

LED control : ID[3] LSM Red LED On

0xFF	0xFF	0x4C	0x53	0x00	0x00	0x00	0x00	0x30	0x05	0x04	0x03	0xfa	0x00	0x00	Checksum
											id 3	250	0	0	

Position control : ID[2] LSM Position '-100' with torqe 80%

0xFF	0xFF	0x4C	0x53	0x00	0x00	0x00	0x00	0x30	0x03	0x04	0x02	0x50	0xff	0x9c	Checksum
											id 2	torq 80	position -100		

Untitled_1 +

File

Edit

Connection

View

Window

Help

New

Open

Save

Connect

Disconnect

Clear Data

Options

View ASCII

Help

0000	FF	FF	4C	53	00	00	00	00	30	0E	04	00	01	01	00	44	OOLS....0.....D
0010	FF	FF	4C	53	00	00	00	00	30	0E	04	00	00	01	00	43	OOLS....0.....C
0020	FF	FF	4C	53	00	00	00	00	30	0E	04	00	01	02	00	45	OOLS....0.....E
0030	FF	FF	4C	53	00	00	00	00	30	0E	04	00	00	02	00	44	OOLS....0.....D
0040	FF	FF	4C	53	00	00	00	00	30	0E	04	00	01	03	00	46	OOLS....0.....F
0050	FF	FF	4C	53	00	00	00	00	30	0E	04	00	00	03	00	45	OOLS....0.....E
0060	FF	FF	4C	53	00	00	00	00	30	0E	04	00	01	04	00	47	OOLS....0.....G
0070	FF	FF	4C	53	00	00	00	00	30	0E	04	00	00	04	00	46	OOLS....0.....F
0080	FF	FF	4C	53	00	00	00	00	30	0E	04	00	01	05	00	48	OOLS....0.....H
0090	FF	FF	4C	53	00	00	00	00	30	0E	04	00	00	05	00	47	OOLS....0.....G
00A0	FF	FF	4C	53	00	00	00	00	30	0E	04	00	01	06	00	49	OOLS....0.....I
00B0	FF	FF	4C	53	00	00	00	00	30	0E	04	00	00	06	00	48	OOLS....0.....H
00C0	FF	FF	4C	53	00	00	00	00	30	0E	04	00	01	07	00	4A	OOLS....0.....J
00D0	FF	FF	4C	53	00	00	00	00	30	0E	04	00	00	07	00	49	OOLS....0.....I
00E0	FF	FF	4C	53	00	00	00	00	30	0E	04	00	01	08	00	4B	OOLS....0.....K
00F0	FF	FF	4C	53	00	00	00	00	30	0E	04	00	00	08	00	4A	OOLS....0.....J
0100	FF	FF	4C	53	00	00	00	00	30	0E	04	00	01	09	00	4C	OOLS....0.....L

눌렀을때

놓았을때 항상 쌍으로 들어 오게 된다.

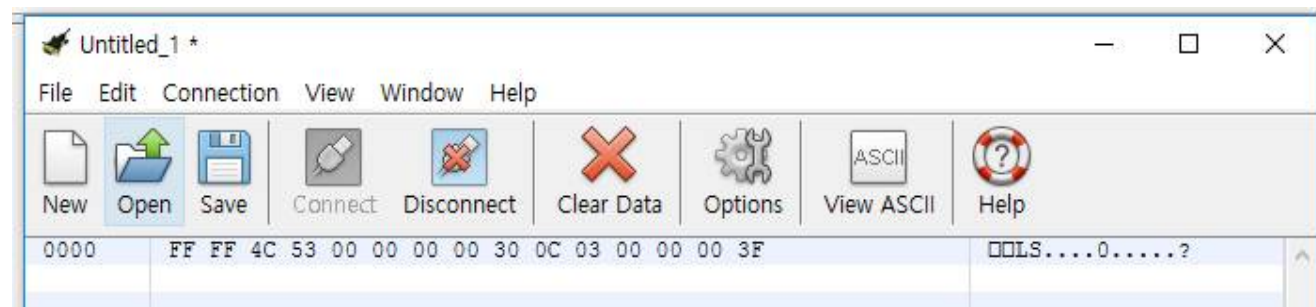
FF FF	헤더	
4C 53	헤더	
00 00	소스주소	
00 00	복사위치주소	
30	exe Type	
0E	exe Remocon	
04	파라메타수	
00~06	모드번호	
01 or 00	Push/Release	
0x00~0x09	왼쪽키값	RF1 눌렀을때 0x0B부터 임
0x00~0x09	오른쪽버튼값	LF1 눌렀을때 0x0B부터 임

0110	FF FF 4C 53 00 00 00 00 30 0E 04 00 00 09 00 4B	00LS....0.....K
0120	FF FF 4C 53 00 00 00 00 30 0E 04 00 01 00 00 43	00LS....0.....C
0130	FF FF 4C 53 00 00 00 00 30 0E 04 00 01 01 00 44	00LS....0.....D
0140	FF FF 4C 53 00 00 00 00 30 0E 04 00 00 01 00 43	00LS....0.....C
0150	FF FF 4C 53 00 00 00 00 30 0E 04 00 01 02 00 45	00LS....0.....E
0160	FF FF 4C 53 00 00 00 00 30 0E 04 00 00 02 00 44	00LS....0.....D
0170	FF FF 4C 53 00 00 00 00 30 0E 04 00 01 03 00 46	00LS....0.....F
0180	FF FF 4C 53 00 00 00 00 30 0E 04 00 00 03 00 45	00LS....0.....E
0190	FF FF 4C 53 00 00 00 00 30 0E 04 00 01 04 00 47	00LS....0.....G
01A0	FF FF 4C 53 00 00 00 00 30 0E 04 00 00 04 00 46	00LS....0.....F
01B0	FF FF 4C 53 00 00 00 00 30 0E 04 00 01 05 00 48	00LS....0.....H
01C0	FF FF 4C 53 00 00 00 00 30 0E 04 00 00 05 00 47	00LS....0.....G
01D0	FF FF 4C 53 00 00 00 00 30 0E 04 00 01 06 00 49	00LS....0.....I
01E0	FF FF 4C 53 00 00 00 00 30 0E 04 00 00 06 00 48	00LS....0.....H
01F0	FF FF 4C 53 00 00 00 00 30 0E 04 00 01 08 00 4B	00LS....0.....K
0200	FF FF 4C 53 00 00 00 00 30 0E 04 00 00 08 00 4A	00LS....0.....J
0210	FF FF 4C 53 00 00 00 00 30 0E 04 00 01 09 00 4C	00LS....0.....L
0220	FF FF 4C 53 00 00 00 00 30 0E 04 00 00 09 00 4B	00LS....0.....K
0230	FF FF 4C 53 00 00 00 00 30 0E 04 00 00 00 00 42	00LS....0.....B
0240	FF FF 4C 53 00 00 00 00 30 0E 04 00 01 00 01 44	00LS....0.....D
0250	FF FF 4C 53 00 00 00 00 30 0E 04 00 00 00 01 43	00LS....0.....C
0260	FF FF 4C 53 00 00 00 00 30 0E 04 00 01 00 02 45	00LS....0.....E
0270	FF FF 4C 53 00 00 00 00 30 0E 04 00 00 00 02 44	00LS....0.....D
0280	FF FF 4C 53 00 00 00 00 30 0E 04 00 01 00 03 46	00LS....0.....F
0290	FF FF 4C 53 00 00 00 00 30 0E 04 00 00 00 03 45	00LS....0.....E
02A0	FF FF 4C 53 00 00 00 00 30 0E 04 00 01 00 04 47	00LS....0.....G
02B0	FF FF 4C 53 00 00 00 00 30 0E 04 00 00 00 04 46	00LS....0.....F
02C0	FF FF 4C 53 00 00 00 00 30 0E 04 00 01 00 05 48	00LS....0.....H
02D0	FF FF 4C 53 00 00 00 00 30 0E 04 00 00 00 05 47	00LS....0.....G
02E0	FF FF 4C 53 00 00 00 00 30 0E 04 00 01 00 06 49	00LS....0.....I
02F0	FF FF 4C 53 00 00 00 00 30 0E 04 00 00 00 06 48	00LS....0.....H
0300	FF FF 4C 53 00 00 00 00 30 0E 04 00 01 00 07 4A	00LS....0.....J
0310	FF FF 4C 53 00 00 00 00 30 0E 04 00 00 00 07 49	00LS....0.....I
0320	FF FF 4C 53 00 00 00 00 30 0E 04 00 01 00 08 4B	00LS....0.....K
0330	FF FF 4C 53 00 00 00 00 30 0E 04 00 00 00 08 4A	00LS....0.....J
0340	FF FF 4C 53 00 00 00 00 30 0E 04 00 01 00 09 4C	00LS....0.....L
0350	FF FF 4C 53 00 00 00 00 30 0E 04 00 00 00 09 4B	00LS....0.....K
0360	FF FF 4C 53 00 00 00 00 30 0E 04 00 01 00 00 43	00LS....0.....C
0370	FF FF 4C 53 00 00 00 00 30 0E 04 00 01 00 0B 4E	00LS....0.....N
0380	FF FF 4C 53 00 00 00 00 30 0E 04 00 00 00 0B 4D	00LS....0.....M
0390	FF FF 4C 53 00 00 00 00 30 0E 04 00 01 00 0C 4F	00LS....0.....O
03A0	FF FF 4C 53 00 00 00 00 30 0E 04 00 00 00 0C 4E	00LS....0.....N
03B0	FF FF 4C 53 00 00 00 00 30 0E 04 00 01 00 0D 50	00LS....0.....P
03C0	FF FF 4C 53 00 00 00 00 30 0E 04 00 00 00 0D 4F	00LS....0.....O
03D0	FF FF 4C 53 00 00 00 00 30 0E 04 00 01 00 0E 51	00LS....0.....Q
03E0	FF FF 4C 53 00 00 00 00 30 0E 04 00 00 00 0E 50	00LS....0.....P
03F0	FF FF 4C 53 00 00 00 00 30 0E 04 00 01 00 0F 52	00LS....0.....R
0400	FF FF 4C 53 00 00 00 00 30 0E 04 00 00 00 0F 51	00LS....0.....Q
0410	FF FF 4C 53 00 00 00 00 30 0E 04 00 01 00 10 53	00LS....0.....S

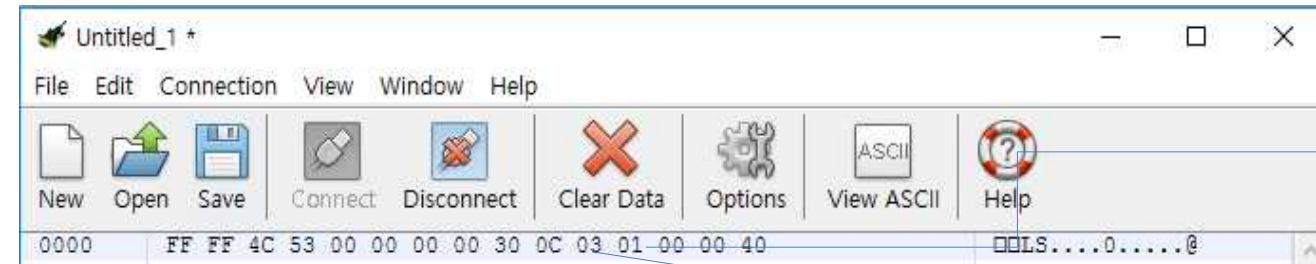
0420	FF FF 4C 53 00 00 00 00 30 0E 04 00 00 00 10 52	00LS....0.....R
0430	FF FF 4C 53 00 00 00 00 30 0E 04 00 01 00 11 54	00LS....0.....T
0440	FF FF 4C 53 00 00 00 00 30 0E 04 00 00 00 11 53	00LS....0.....S
0450	FF FF 4C 53 00 00 00 00 30 0E 04 00 01 00 12 55	00LS....0.....U
0460	FF FF 4C 53 00 00 00 00 30 0E 04 00 00 00 12 54	00LS....0.....T
0470	FF FF 4C 53 00 00 00 00 30 0E 04 00 01 00 13 56	00LS....0.....V
0480	FF FF 4C 53 00 00 00 00 30 0E 04 00 00 00 13 55	00LS....0.....U
0490	FF FF 4C 53 00 00 00 00 30 0E 04 00 00 00 00 42	00LS....0.....B

COM4 / 9600 8-N-1
Connected 00:26:31

TX RX RTS CTS DTR DSR DCD RI



Motion Stop

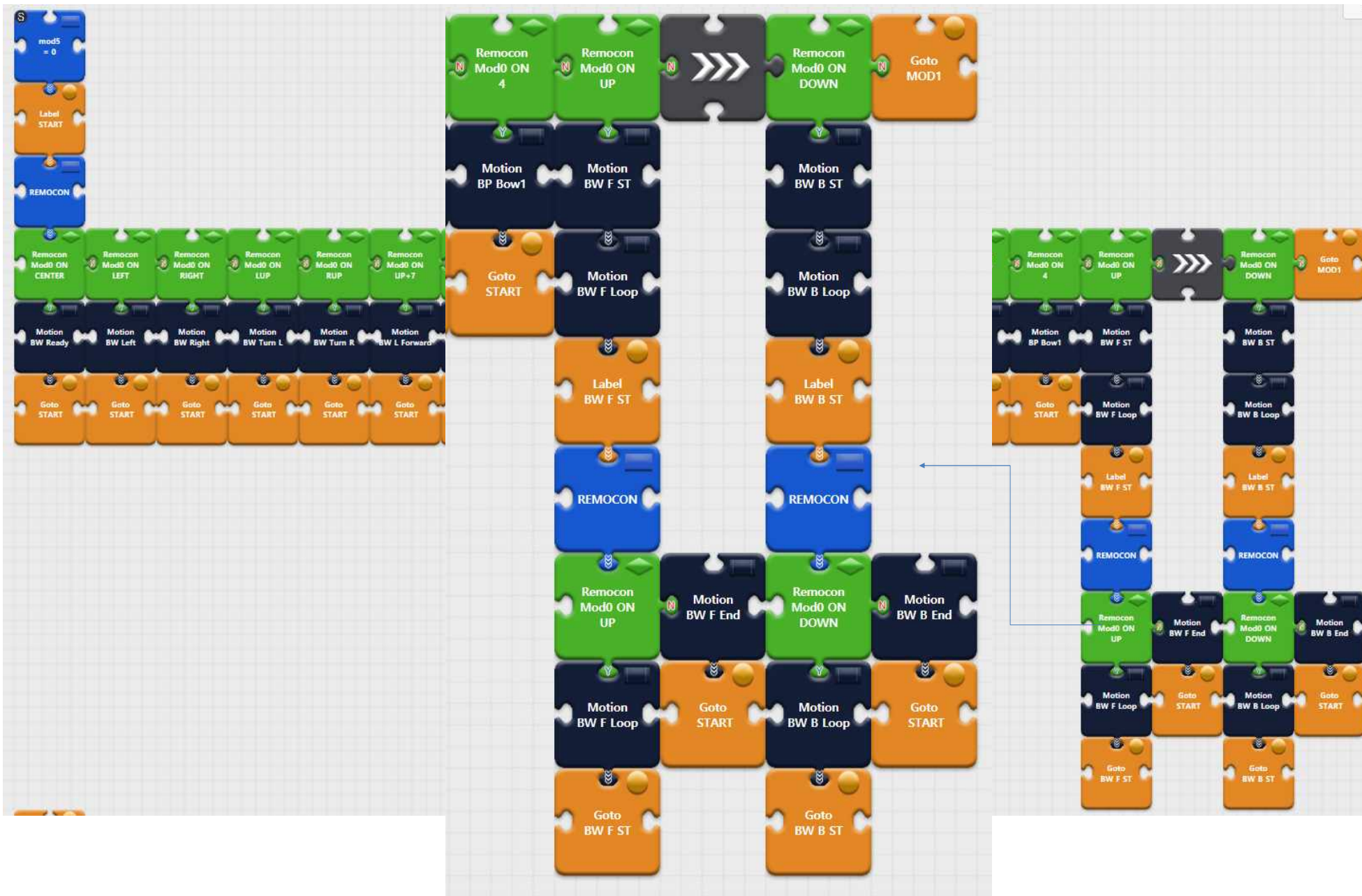


Music Mode 1

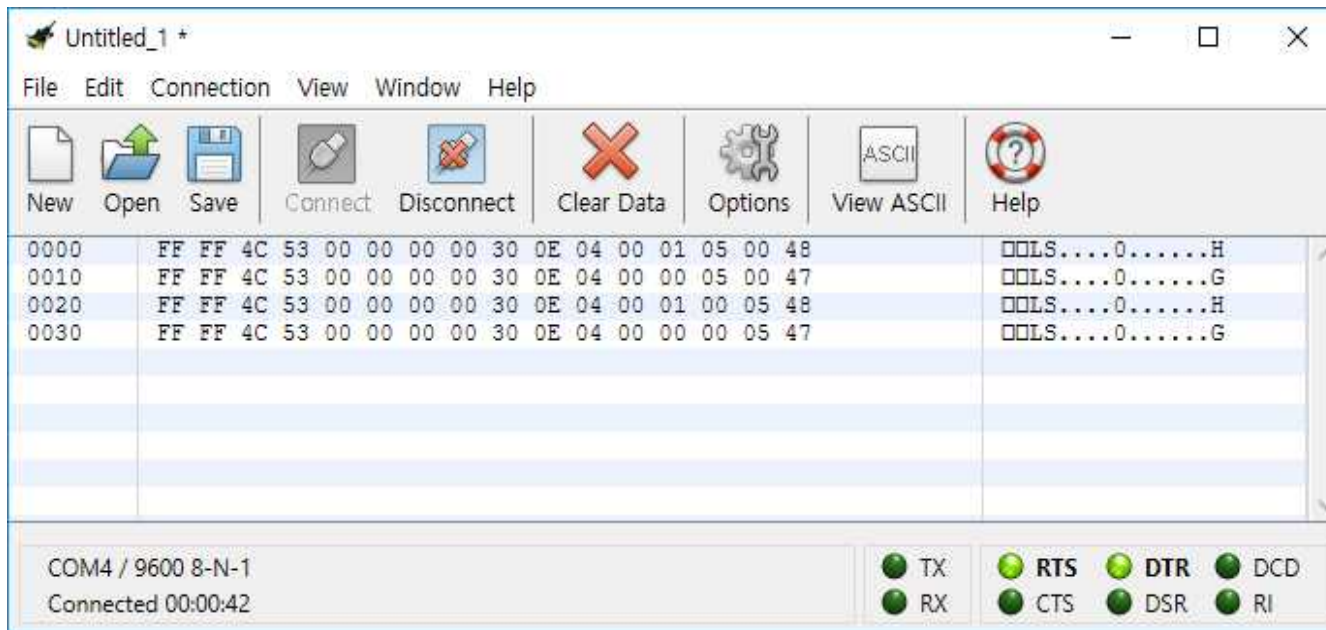
댄스번호 1~6

파라메타수

exe motion



실제 리모콘에서 보내는 내용은 이렇다



이것을 컨트롤러는 받아서 해석해서 사용해야 한다.

240 전원 끄기
241 전원 켜기
242 모드 바꾸기
243 Led 램프 바꾸는것인가? 잘 모르겠음 나중에 수정하겠음